

07.1

Describe how an ATP molecule is formed from its component molecules.

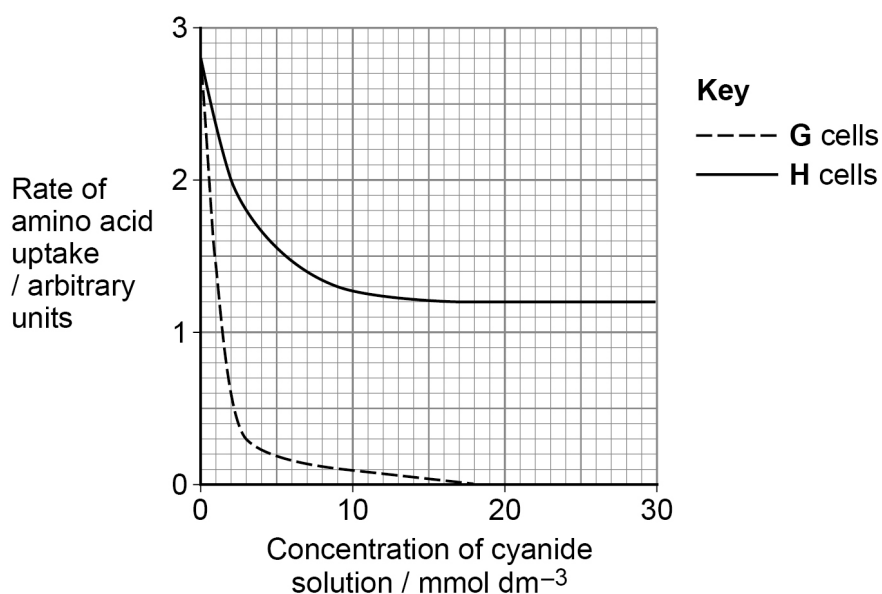
[4 marks]

A scientist investigated the effect of cyanide on the rate of amino acid uptake in two types of *Escherichia coli*, **G** and **H**.

- **G** cells produce enzymes involved in ATP production **only** on their cell-surface membrane.
- **H** cells produce enzymes involved in ATP production on their cell-surface membrane **and** in their cytoplasm.

Figure 8 shows her results.

Figure 8



07.2

Use **Figure 8** to calculate the percentage decrease in the rate of amino acid absorption by **H** cells in 30 mmol dm^{-3} cyanide solution.

[1 mark]

Answer _____ %

07.3

Using **Figure 8** and the information provided, what can you conclude about amino acid uptake by **G** cells and by **H** cells?

[3 marks]

8

Turn over for the next question

Turn over ►

